Bonds Market

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January 9, 2024

Outline

1. Bond Market
2. Government and Corporate Bonds
3. Discounted value of cash flows
4. Term Structure
5. Price-Yield Conventions
6. References

Definition (Bond)

A bond is a debt instrument issued by borrowers to obtain cash for either short term or long term needs.

Example (Issures of bonds)

- Sovereign governments
- State governments
- Corporations

Example (Buyers of bonds)

- Pension funds
- Insurance companies
- Foreign countries
- Individual investors

Purpose of issuing and buying bonds

Purpose of issuers

- Governments issue bonds to generate funds for constructing roads, expanding railways, spending on defense, providing education and for other expanses.
- Financial institutions issue bonds to invest in higher yielding assets and for expanding the business.

Purpose of buyers

- The main reason of investing in bonds is to have a diversify portfolio of investment.

Example (10 year US Treasury note)

1. Pricing date — July 22, 2005
2. Price — 99.214
3. Face value — 100
4. Market yield — 4.223%
5. Issuer — US Treasury
6. Maturity date — May 15, 2015
7. Coupon — 4.125%
8. Issued amount — $ 24.27 billion
9. Outstanding amount — $ 22 billion
10. Issue date — May 15, 2005

Example (General Motors bond)

1	Pricing date	Dec 29, 2005
2	Price	66
3	Market yield	13.299%
4	Issuer	General Motors
5	Maturity date	July 15, 2023
6	Coupon	8.25%
7	Issued amount	$ 1.25 billion
8	Outstanding amount	$ 1.25 billion
9	Issue date	June 26, 2003
10	Call	GM has the option to call
11	Rating	Noninvestment grade

The idea of interest rate is to compensate the lender for his money since money has time value. Thus, Rs 100 today is equivalent to Rs 100+**compensation** in future.

Further, the term **compensation** depends on many factors such as

- Duration of borrowing
- Trustworthiness of the party that is borrowing

India's ancient idea on interest

Chanakya's interest rate structure was risk-weighted. In his opinion if a person was doing business from safe way(less risk) charge him/her less and if he/she was doing business from forest or sea(risk increased) charge him/her more. In other words the rate of interest increased with the risk involved in the borrowers' business.

Cash-flows from a bond

1. Suppose, there is a bond with face value \$100 and coupon rate C and having a maturity of 1 year.
2. Suppose the coupon payment frequency is **semi-annual**.
3. The cash-flows from this bond include a coupon payment of $\$C/2$ after 6-months and a payment of $\$100 + C/2$ after 1-year.
4. If the coupon payment frequency is **quarter**.
5. In this case the cash-flows from this bond include a first coupon payment of $\$C/4$ after 3-months, a second coupon payment of $\$C/4$ after 6-months, a third coupon payment of $\$C/4$ after 9-months and a final payment of $\$100 + C/4$ after 1-year.

Discounted value of a cash-flow

- As we discussed earlier, Rs 100 received today is different than receiving Rs 100 after one year. In fact if prevailing one-year interest rate is 2% the value of Rs 100 to be received a year later will have value only Rs $100/1.02 = 98.04$ today.
- In other words if we have Rs 98.04 today we can invest this amount in bank and will receive Rs $98.04(1 + .02) = Rs100$ a year later. This Rs 98.04 today is called the **present value(PV)** of the Rs 100 a year later.
- The concept of **present value** is used extensively in finance for comparing two different cash-flows.

Discounted value of a bond cash-flow

- Suppose, there is a bond with face value \$100 and coupon rate C% and having a maturity of 5 years.
- Suppose the coupon payment frequency is **annual**.
- The cash-flows from this bond include a first coupon payment of \$$C$ after 1-year, a second coupon payment of \$$C$ after 2-year, a third coupon payment of \$$C$ after 3-year, a fourth coupon payment of \$$C$ after 4-year and a final payment of \$$100 + C$ after 5-year.
- Further, suppose the interest rate is r% **flat** and compounding is **simple**.
- the present value of the bond is given by

$$PV = \frac{C}{1+r} + \frac{C}{(1+r)^2} + \frac{C}{(1+r)^3} + \frac{C}{(1+r)^4} + \frac{100+C}{(1+r)^5} \tag{1}$$

Convertible Bonds

A **Convertible Bond** is a hybrid fixed-income security that yields interest payments, but can also be converted into a predetermined number of **Common Stocks**.

- The price of convertible bond is sensitive to changes in interest rates.
- The **conversion ratio** determines how many shares you can get by converting one bond.
- If the stock price increase significantly, the investor can convert the bond to stock and hold or sell the stock whenever deem appropriate.

Source of Risks for Bond Holders

Interest rate risk (Market risk)

- This is the major factor that affects bond prices
- The price of bond changes in the opposite direction of interest rate change
- All bonds are exposed to interest rate risk

Inflation risk

- Inflation reduces purchasing power that mean same amount of money has lesser value in period of high inflation.
- In other words same bond has different yields(real) in different inflation scenarios

Credit risk

- Inability of issuer to pay coupon and/or principal. For example
  - Corporate
  - Emerging market
  - High-yield bonds

Liquidity risk

- Inability to close position without substantial costs
- Some examples of illiquid bonds include municipal, corporate, and emerging market bonds

Term structure of interest rates (or yield curve)

Definition (Yield curve)

A yield curve is the relationship between spot rates and term to maturity. It is a measure of the market's expectations of future interest rates given the current market conditions.

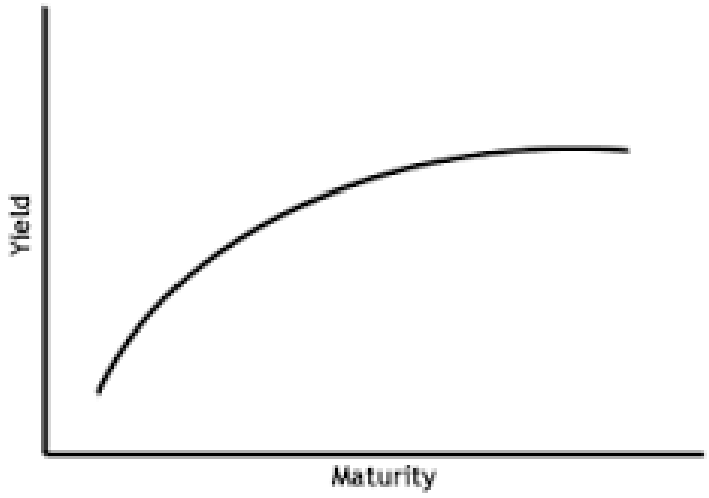

Figure: Normal Yield Curve

Credit spread Treasury and corporate bond

Since a Treasury bond is riskless and a corporate bond carry substantial credit risk, investors demand additional yield from corporate bonds. This yield difference is called **credit spread**.

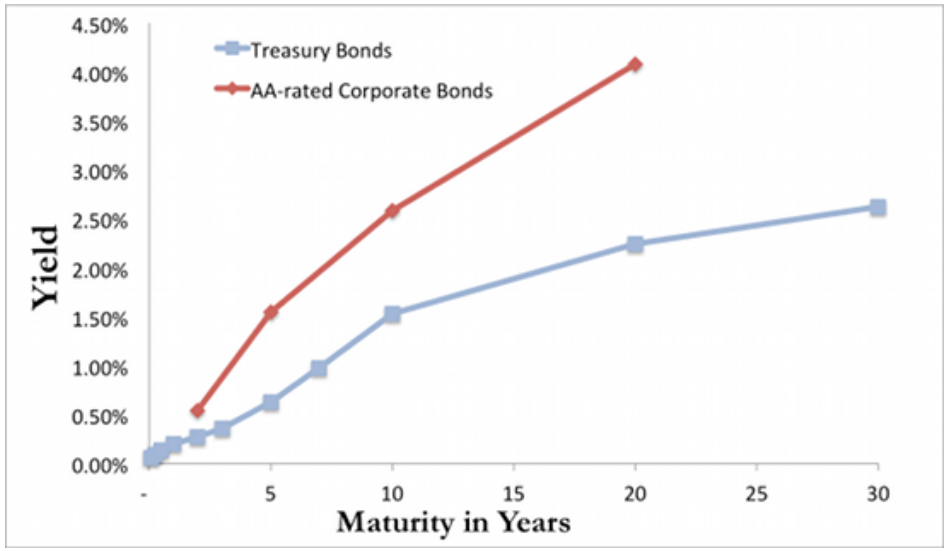

Figure: Credit Spread

Future Values

- The future value (FV) of investing P today for N years at an annual interest rate r is

$$FV = P(1+r)^N.$$

- If the interest is compounded semiannually, that is, we are paid interest semiannually and we earn interest on the interests, then the future value after N years will be

$$FV = P\left(1+\frac{r}{2}\right)^{2N}.$$

- Proceeding this way, the future value of P invested today for N years with (m compounding intervals per year) is

$$FV = P\left(1+\frac{r}{m}\right)^{Nm}.$$

- In the limiting case as $m \to \infty$, we have $FV = Pe^{rN}$ which is also called **continuous compounding**.

A security that pays C per period for N periods is known as **annuity.** Suppose the annual interest rates are r, then

$$\begin{aligned} FV &= C(1+r)^{N-1} + C(1+r)^{N-2} + C(1+r)^{N-3} + \cdots + C(1+r) + C \\ &= C\left(1 + (1+r) + (1+r)^2 + \cdots + (1+r)^{N-1}\right) \\ &= \frac{C}{r}\left((1+r)^N - 1\right). \end{aligned}$$

Example

Consider a loan in which a payment of $C = 100$ per annum has to be made for the next 10 years. Let the interest rate is 9%. What is the future value of this annuity ?

The future value is

$$FV = \frac{100}{0.09}\left[(1+0.09)^{10} - 1\right] = 1519.29.$$

Present Values

- For investors, having securities, which promise cash flows at future dates, the concept of **present value (PV)** is important.
- The PV of \$100 to be received after N years from today with annual compounding is
$$PV = \frac{100}{(1+r)^N}.$$
- Similarly, the PV of \$100 to be received after N years from today with m interest compounding intervals per year is
$$PV = \frac{100}{\left(1+\frac{r}{m}\right)^{Nm}}.$$
- With continuous compounding it becomes
$$PV = 100e^{-rN}.$$

Example

Consider two zero coupon bonds A and B with maturity 1 and 2 years respectively. Both bonds have a face value of Rs. 1000. The one year and two year interest rates are 8% and 10% respectively. Find the present values of bonds A and B.

Solution: We have

$$PV_A = \frac{1000}{1.08} = Rs.925.93.$$

Similarly,

$$PV_B = \frac{1000}{1.10^2} = Rs.826.45.$$

Example

Suppose the spot rates for one year and two years are 8% and 10% respectively. Find the price of a 5% coupon bond with maturity two years. The bond pays annual coupons with face value Rs. 1000.

Solution: We have

$$PV = \frac{50}{1+0.08} + \frac{1050}{(1+0.10)^2} = Rs.914.06.$$

The PV of an annuity which pays C for N years with annual compounding is

$$\begin{aligned} PV &= \frac{C}{(1+r)} + \frac{C}{(1+r)^2} + \frac{C}{(1+r)^3} + \cdots + \frac{C}{(1+r)^N} \\ &= \frac{C}{1+r}\left[1 + \frac{1}{1+r} + \cdots + \frac{1}{(1+r)^{N-1}}\right] \\ &= \frac{C}{1+r}\frac{1-(1+r)^{-N}}{1-\frac{1}{(1+r)}} \\ &= \frac{C}{r}\left[1-(1+r)^{-N}\right]. \end{aligned}$$

Example

Consider an annuity of \$100 per year for the next 10 years at an interest rate of 9%. What is the PV of this annuity ?

$$PV = \frac{100}{0.09}\left[1-(1+0.09)^{-10}\right] = \$641.77.$$

Yield to Maturity (YTM) or Internal Rate of Return

The YTM or IRR denoted by y, is the rate of discount at which the present value of the promised future cash flows equals the price of the security.

- The price P of a bond that pays annual dollar coupons of C for N years, per \$100 of face value is
$$P = \frac{C}{1+y} + \frac{C}{(1+y)^2} + \cdots + \frac{C+100}{(1+y)^N}.$$
- Given the market price P, we can find out the bond's YTM from this equation.
- By taking $C = 100c$, we have
$$P = \frac{100c}{y}\left[1-(1+y)^{-N}\right] + \frac{100}{(1+y)^N}.$$
- There is an important consequence of the formula: that is when the coupon rate is equal to the YTM, we have $P = 100$, i.e. the asset trade at par.
- If $c > y$, then $P > 100$, i.e. its trade at premium to par.
- If $c < y$, then $P < 100$, i.e. its trade at discount to par.

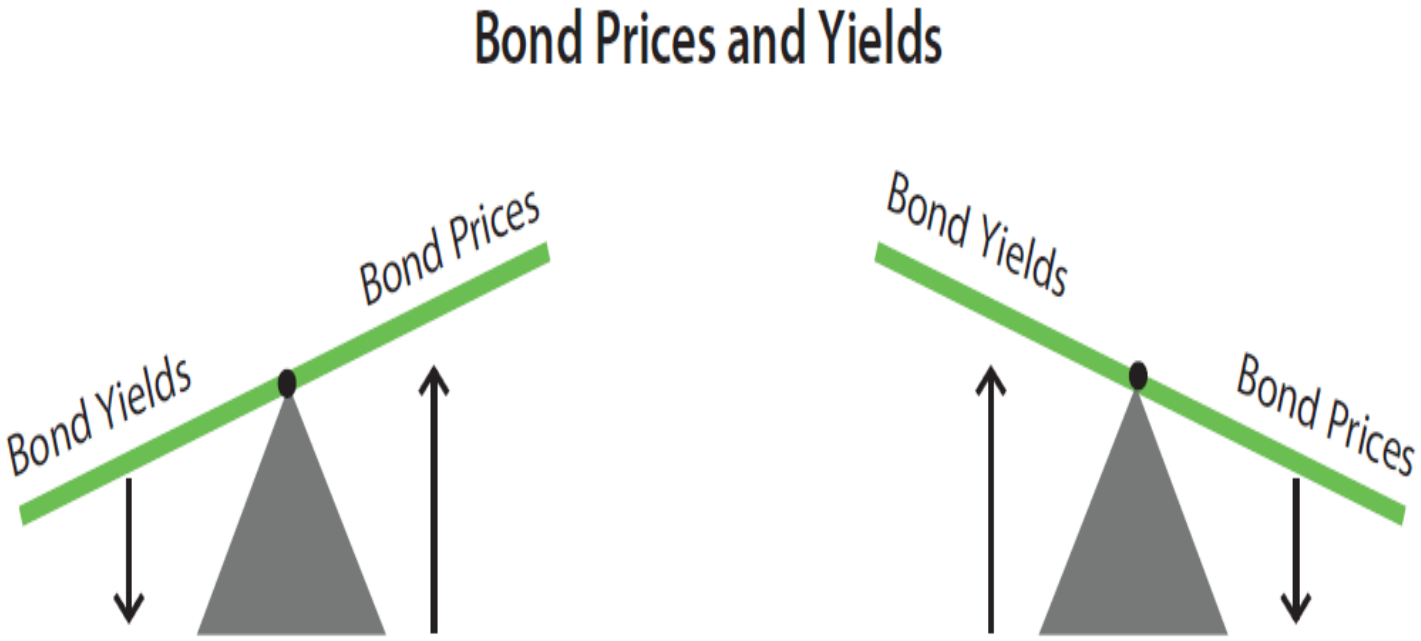

Figure: Price moves in opposite direction of yield

Example

The price of a 5% coupon bond with maturity two years is Rs. 914.06. The bond pays annual coupon with face value Rs. 1000. Find the YTM.

Solution: We have

$$\frac{50}{1+y} + \frac{1050}{(1+y)^2} = Rs.914.06.$$

This implies $y = 9.95\%$.

Term Structure or Spot Rates Curve

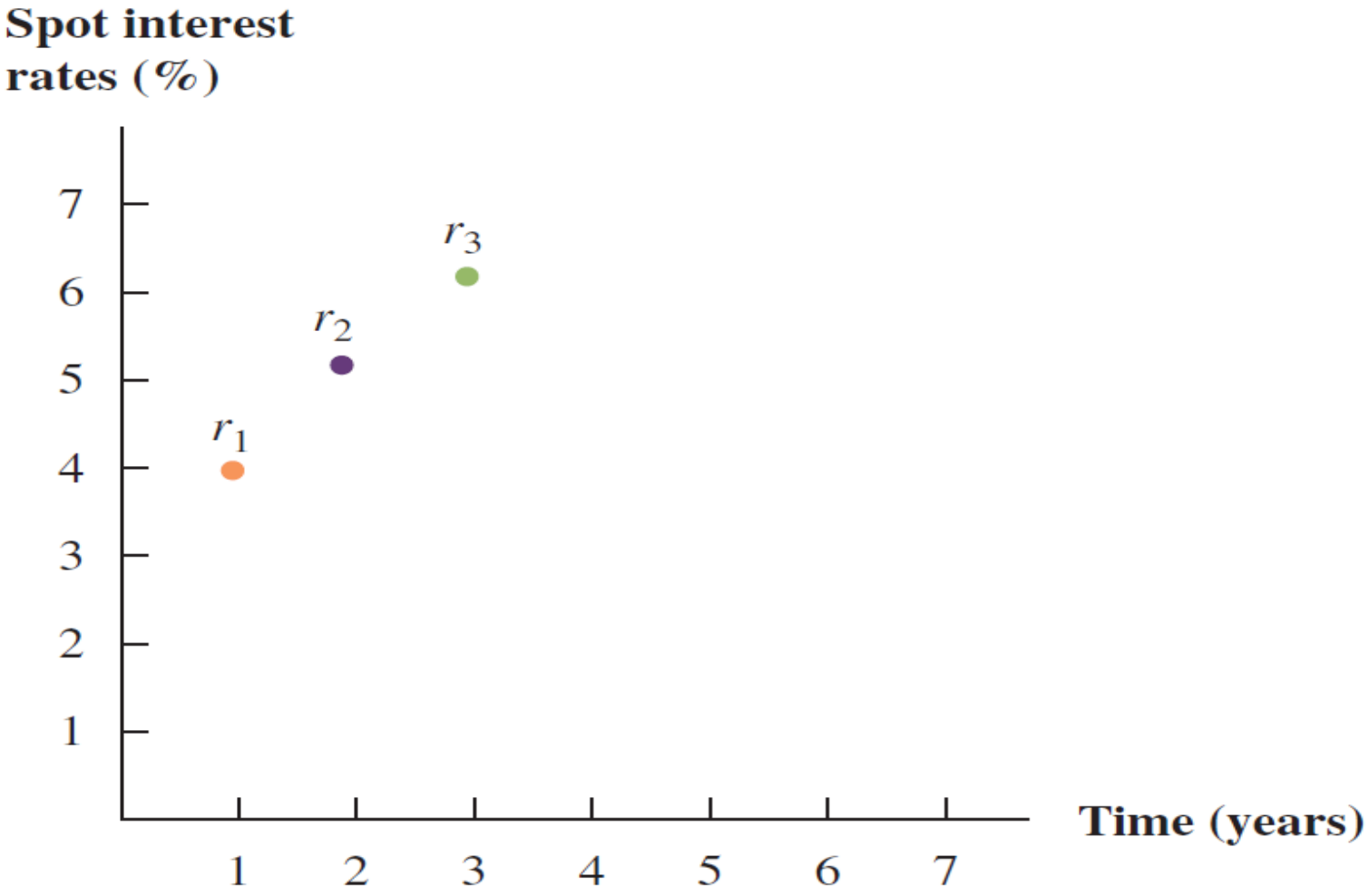

Figure: Spot Rates

US Treasury Yield Curve

Date	1 Mo	2 Mo	3 Mo	4 Mo	6 Mo	1 Yr	2 Yr	3 Yr	5 Yr	7 Yr	10 Yr	20 Yr	30 Yr
01/02/2024	5.55	5.54	5.46	5.41	5.24	4.80	4.33	4.09	3.93	3.95	3.95	4.25	4.08
01/03/2024	5.54	5.54	5.48	5.41	5.25	4.81	4.33	4.07	3.90	3.92	3.91	4.21	4.05
01/04/2024	5.56	5.48	5.48	5.41	5.25	4.85	4.38	4.14	3.97	3.99	3.99	4.30	4.13
01/05/2024	5.54	5.48	5.47	5.41	5.24	4.84	4.40	4.17	4.02	4.04	4.05	4.37	4.21
01/08/2024	5.54	5.48	5.49	5.39	5.24	4.82	4.36	4.11	3.97	3.99	4.01	4.33	4.17

Figure: The inverted US Treasury Curve, which is an harbinger of slow down in the market.

Suppose the spot rates for one year and two years are given by r_1 and r_2 respectively. We can determine the forward rate f_2 as follows

$$(1 + r_2)^2 = (1 + r_1)(1 + f_2).$$

Which gives

$$f_2 = \frac{(1 + r_2)^2}{(1 + r_1)} - 1.$$

In general

$$f_n = \frac{(1 + r_n)^n}{(1 + r_{n-1})^{n-1}} - 1,$$

where f_n is the forward rate over the nth year, r_n and r_{n-1} are the spot rates for n and $n - 1$ years.

Example

If the one year and two-year spot rates are 7 and 12 percent respectively, find f_2.

$$f_2 = \frac{1.12^2}{1.07} - 1 = 17.23\%.$$

Forward rates example

Example

Suppose the spot rates are $r_1 = 5\%$, $r_2 = 6\%$, $r_3 = 7\%$ and $r_4 = 6\%$. find the forward rates for 2nd, 3rd and 4th years.

Solution: We have

$$f_2 = \frac{1.06^2}{1.05} - 1 = 7.01\%$$

$$f_3 = \frac{1.07^3}{1.06^2} - 1 = 9.03\%$$

$$f_4 = \frac{1.06^4}{1.07^3} - 1 = 3.06\%.$$

In bond quotes $+$ denotes $\frac{1}{64}th$ and $++$ denotes $\frac{1}{128}th$.

Table 2.3 Treasury Quotes as of June 26, 2008

Maturity in Years	Quoted Price (in Units of 32nds)	Computed Price (Decimals)
2	100-13+	100.421875
5	100-15+	100.484375
10	98-22	98.687500
30	96-10+	96.312500

Source: Bloomberg.

Figure: Bond prices in practice

Quoted Price and the Invoice Price for Treasury Bonds/Notes

- For treasury notes (maturity between 2 and 10 years) and treasury bonds (maturity 30 years), the **quoted price** also called clean price, flat price is typically are not the **invoice price** (or the dirty price).
- To come to the invoice price, you need to add the **accrued interest** (AI) to the flat price.
- AI is the coupon income that accrues from the last coupon date to the settlement date of the transaction.
- This amount the buyer of the bond to the seller to receive the full coupon on next coupon date.

Example

The quotations for June 26, 2008, of a U.S. Treasury note with a 3.875% coupon and a maturity date of May 15, 2018, appear in Figure 2.4. Recall that the price quotations are in 32nds.

Maturity date	Coupon	Quoted clean price in 32nds	Quoted clean price in decimals
5/15/2018	3.875%	98-22	98.6875

Figure: Bond clean and dirty price

- The last coupon date(LCD), or the dated date (when the first coupon starts to accrue), which in this case is May 15, 2008. The next coupon date (NCD) is November 15, 2008. So, the number of days between the NCD and the LCD is 184 days.
- The number of days between the last coupon date and the settlement date, June 27, 2008, is 43 days. The accrued interest AI is

$$AI = \frac{43}{184} \times \frac{3.875}{2} = 0.452785.$$

- Then the invoice (dirty) price is 98.6875 + 0.452785 = 99.140285.

Zero coupon bond and spot yield curve

A zero coupon bond is a different kind of bond in which the face value is repaid at the time of maturity. It does not make periodic interest payments or have so-called coupons, hence the term zero coupon bond.

- Suppose there is a zero coupon bond with face value 100 has current price P with a maturity of t years. Then the spot yield y_t satisfies $P(1+y_t)^t = 100$ or $y_t = \left(\frac{100}{P}\right)^{1/t} - 1$.
- If we plot the the yields y_t on y-axis and t on x-axis, we get the spot yield curve (or the zero-coupon yield curve).

Zero coupon yield curve India

Residual Maturity	Yield			ZC Price			
	Last	Chg 1M	Chg 6M	Last	Chg 1M	Chg 6M	Fx
3 months	3.600%	+4.0 bp	+16.0 bp				
6 months	4.030%	+12.0 bp	+43.0 bp				
1 year	4.323%	-3.1 bp	+37.3 bp	95.86	+0.03 %	-0.35 %	
2 years	4.923%	+10.7 bp	+63.7 bp	90.84	-0.20 %	-1.21 %	
3 years	5.319%	+18.8 bp	+63.2 bp	85.60	-0.53 %	-1.79 %	
4 years	5.933%	+21.0 bp	+33.8 bp	79.41	-0.79 %	-1.27 %	
5 years	6.035%	+27.5 bp	+35.4 bp	74.60	-1.30 %	-1.66 %	
6 years	6.390%	+23.3 bp	+29.5 bp	68.96	-1.30 %	-1.65 %	
7 years	6.532%	+24.9 bp	+17.2 bp	64.22	-1.62 %	-1.12 %	
8 years	6.601%	+24.6 bp	+16.7 bp	59.97	-1.83 %	-1.24 %	
9 years	6.505%	+7.3 bp	+29.3 bp	56.71	-0.61 %	-2.46 %	
10 years	6.636%	+22.5 bp	+50.6 bp	52.60	-2.08 %	-4.64 %	
11 years	6.829%	+18.0 bp	+14.4 bp	48.35	-1.85 %	-1.49 %	
12 years	6.955%	+25.8 bp	+11.6 bp	44.63	-2.85 %	-1.28 %	
13 years	7.083%	+27.6 bp	+26.4 bp	41.08	-3.30 %	-3.16 %	
14 years	7.080%	+28.1 bp	+28.8 bp	38.38	-3.62 %	-3.69 %	
15 years	7.053%	+29.4 bp	+15.1 bp	35.98	-4.03 %	-2.10 %	
19 years	7.137%	+22.4 bp	+5.0 bp	26.99	-3.88 %	-0.88 %	
24 years	7.205%	+26.5 bp	+10.1 bp	18.83	-5.76 %	-2.23 %	
30 years	7.274%	+29.4 bp	+12.9 bp	12.17	-7.87 %	-3.49 %	

The role of central bank

- The **central bank** of a country is solely responsible for formulating and implementing the nation's monetary policies.
- Further, it supervises the banking sector and capital markets.
- India - Reserve Bank of India (RBI).
- USA - Federal Reserve or simply the Fed.
- European zone - European Central Bank (ECB)
- Australia - Reserve Bank of Australia.

Functions of RBI

- **Issue of currency notes:** RBI prints the currency notes of various denominations except the one rupee notes (which are issued by MoF).
- **Bankers' Bank:** All the commercial banks are customers of RBI. RBI lends money to all the commercial banks of the country.
- **Controller of the money supply in the economy:** When RBI feels that there is enough money supply in the economy and it may cause an inflationary situation in the country then it squeezes the supply through its **monetary policy**.
- **Custodian of the Foreign Reserves:** For the purpose of keeping the foreign exchange rates stable, the RBI buys and sells foreign currencies.

Monetary Policy of RBI

These methods affect the money supply, cost of money and availability of credits.

- **Bank Rate Policy:** Bank rate is the minimum lending rate of the central bank. When the RBI finds that the inflation has been increasing continuously, it raises the bank rate so the borrowing from RBI becomes costly.

- **Open market operations:** When inflation start rising and there is need to control them, the RBI sells securities.

- **Changes in Reserve Ratios:** The CRR (cash reserve ratio) and SLR (statutory liquidity ratio) are two main deposit ratios, which reduces or increases the cash balance of commercial banks. **CRR** is the percentage of a bank's deposits to be held in the form of cash with RBI. **SLR** is the minimum percentage of deposits that a bank has to maintain in form of gold, cash or other approved securities. These are not reserved with the reserve bank of India but with the bank only.

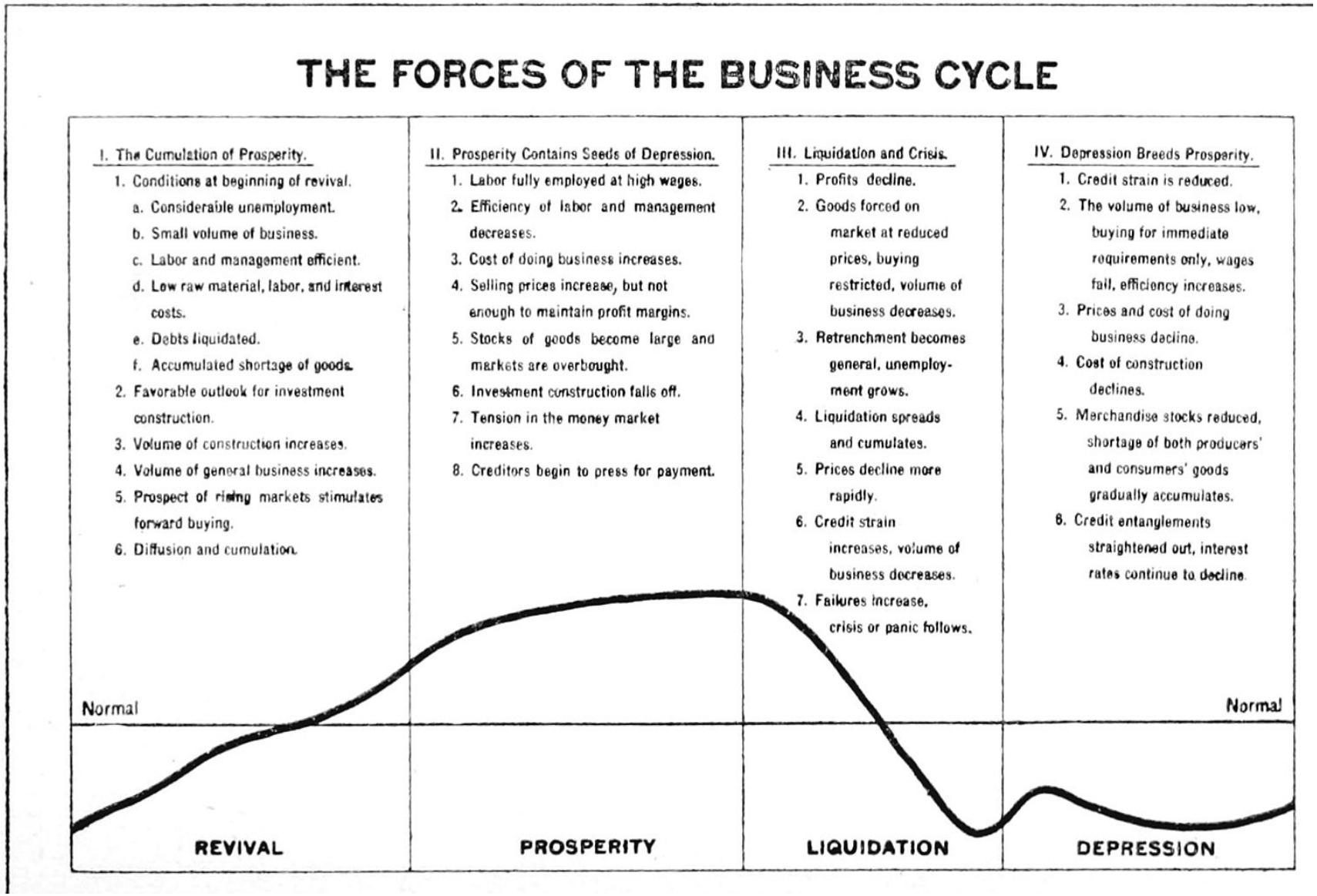

Figure: Forces of Business Cycle

There are many methods which have been used to model the yield curve. We can put them into three categories; spline based models, function based models, and lastly stochastic models.

Spline Based Models

The idea of spline based models is to interpolate a spline function from the pre-known points. To get some features like continuous second derivative (which can be used in analysis of curvature) a cubic spline method is widely used.

Function based Models

The most popular function based models are **Nelson-Siegel**, and **Svensson models**. In fact, Svensson model is an extension of Nelson-Siegel model.

Stochastic Models

The stochastic modeling of yield curve depends on interest rate modeling, especially on short-rate modeling. The idea is to obtain an explicit pricing formula for the zero-coupon bond and then extracting the yield from that. The most used interest rates models are: **Vasicek Model (1977)**, **Cox-Ingersoll-Ross (CIR) Model (1985)** and some others.

Vasicek Model

The Vasicek model describes the short rate dynamics by the following SDE:

$$dr_t = (\theta - ar_t)dt + \sigma dw_t,$$

where θ, $a > 0$ and σ is constant.

Cox-Ingersoll-Ross (CIR) Model

The CIR model describes the short rate dynamics by the following SDE:

$$dr_t = (\theta - ar_t)dt + \sigma\sqrt{r_t}dw_t,$$

where θ, $a > 0$ and σ is constant.

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